# Sahil Kumar

System Engineer (1+ YOE) | Full Stack | AI/ML Engineer

(+91) - 9832857637 | Sahil Kumar | Github | Linkedin | Leetcode

## Work Experience

**System Engineer - Tata Consultancy Services (TCS) | December 2024 - Present**

Tech Stack **: Python, FastAPI, Mongodb, React, Databricks, Git, Gitlab**

- Built a clustering-based recommendation system on **Databricks** for a client proof-of-concept, and designed and hosted a **REST API** endpoint on Databricks for real-time recommendation serving, demonstrating end-to-end ML architecture from model development to production-style deployment
- Contributed to an **AI-powered Image Context Understanding System**, building the **interface layer using LangGraph and React** to enable agentic query workflows on top of multimodal models for semantic scene analysis — helping achieve **92% relevance accuracy** across **10,000+ test images**
- Contributed to a time-series forecasting and anomaly detection pipeline, designing the orchestration logic to automatically route input data to the appropriate task, and collaborating on model validation and benchmarking that **improved accuracy by 20%**
- Wrote clean, testable code with comprehensive unit and integration test coverage.
- Collaborated closely with cross-functional teams including Infra, Security, and Product to align technical solutions with business goals.

## Technical Skills

**Programming Languages :** C, C++, Python, Java, JavaScript
**Backend:** Node.js, FastAPI
**Frontend:** React.js, Redux
**Databases :** MongoDB, MySQL, PostgreSQL, Supabase
**CI/CD & Tools :** Git, GitHub, GitLab, GitHub Actions, Docker
**Cloud :** AWS (EC2, S3, IAM, Lambda, API Gateway, CloudFormation/SAM), Vercel, Databricks (Apps, Compute, Workspace)
**Scripting :** Bash
**Testing & Automation :** Selenium, PyTest, Page Object Model (POM)
**AI/ML :** Deep Learning, LangChain, LangGraph, Agentic AI, Prompt Engineering, RAG, LLM, GenAI
**Other Technical Skills :** Data Structure and Algorithm, Object Oriented Programming, Database Management System, System design, Agile Development

## Projects

**Personalized Budget Tracker | August 2026**

Tech Stack : **Python, FastAPI, SQLAlchemy (async), PostgreSQL (Supabase), Docker, AWS (Lambda, API Gateway, IAM, SAM/CloudFormation), React.js, Vercel**

- Architected and built a full-stack budget tracking application with a layered **FastAPI backend** (router → service → repository) enabling transaction logging, custom percentage-based budget "buckets," and actual-vs-target spending analytics
- Containerized the backend with **Docker** and deployed it as an **AWS Lambda** function behind **API Gateway using AWS SAM/CloudFormation**, eliminating server management while supporting serverless auto-scaling
- Integrated S**upabase-hosted PostgreSQL** via a pooled async connection (SQLAlchemy + asyncpg), designing a normalized schema with foreign-key and check constraints to enforce data integrity at the database level
- Implemented JWT-based authentication, idempotent transaction creation, and typed exception handling mapped to appropriate HTTP status codes for robust API error handling
- Built a responsive **React (Vite)** frontend with Recharts-based spending trend and budget allocation visualizations, deployed on Vercel with GitHub-integrated CI/CD
- Configured IAM roles and cross-service permissions across Lambda, API Gateway, and ECR to support the containerized deployment pipeline

## Education

**National Institute of Technology , Durgapur** | B.tech in Computer Science Engineering - 7.8 CGPA | 2020 - 2024
**Kendriya Vidyalaya Sangathan , Chittaranjan** | 12th Board (CBSE) - 90.8% | 2020
**Kendriya Vidyalaya Sangathan , Chittaranjan** | 10th Board (CBSE) - 75.8% | 2018

## Achievements

- Solved 600+ Data Structures & Algorithms problems on LeetCode
- Achieved a peak LeetCode Contest Rating of 1561 (Top ~20%)
- Top 5% in JEE Mains 2020
- 12th CBSE Board school topper (90.8%)